Available online at www.sciencedirect.com

ScienceDirect

Procedia CIRP 81 (2019) 558–563

www.elsevier.com/locate/procedia

52nd CIRP Conference on Manufacturing Systems

Study on AGVs battery charging strategy for improving utilization

Xiangnan Zhan, Liyun Xu*, Jian Zhang, Aiping Li

Institute of Advanced Manufacturing Technology, Tongji University, Shanghai 201804, P.R.China

* Corresponding author. Tel.: +86-21-69589750; fax: +86-21-69589750. *E-mail address:* Lyxu@tongji.edu.cn

Abstract

The battery charging strategy of automation guided vehicles (AGVs) is playing an important role in improving the utilization of AGVs. This paper proposes two dual-stage battery charging strategies of AGVs, which equip with lithium-ion battery for improving the utilization. In stage 1, two routing decisions based on heuristic algorithm are developed, namely the nearest charging station (NCS) and the minimum delay charging station (MDCS); while in stage 2, the duration of each charging occurrence is reduced considering the charging characteristic of lithium-ion battery. A real case is adopted to illustrate the applicability and effectiveness of the proposed approach.

Keywords: automated guided vehicles; utilization; battery charging strategy.

1. Introduction

Automated guided vehicles (AGVs) are unmanned vehicles that can automatically navigate based on machine vision, lasers or wires in the floor. Thank to the rapidity flexibility and efficiency, they are commonly used for materials horizontal movement in manufacturing system and warehouses. According to the research of Schmidt et al. [1], AGVs used battery as their source of energy, had several economic and technical advantages. Automatic charging and battery swapping are two familiar patterns for battery charging of AGVs [2]. This paper focuses on the application of AGVs with automatic charging in manufacturing system.

Some manufacturing factories need to improve their manufacturing capacity in the short term to meet the increasing market demand. This phenomenon happens when rush orders occur or the product forecasts are underestimated. At present, scholars have proposed a variety of methods for increasing the production of manufacturing system with AGVs. These methods can be divided into three categorie: 1. add new AGVs, 2. optimize the facility layout, 3. improve the utilization of single AGV.

By adding new AGVs, more output can be expected, meanwhile, the job shop scheduling problem considering the congestion and the job transportation times must be addressed [3]. Ravi Kumar el at. [4] proposed a heuristic approach to optimize the facility layout that not only increased the production, but also optimized both material handling costs and rearrangement costs of manufacturing multi-products. But it is impossible within a short time and a large investment is needed. In the case of improving AGVs utilization, Yang proposed a dynamic time estimation based on AGV dispatching algorithm to increase the production [5]. In summary, one of the system potential bottlenecks for improving manufacturing capacity is the working time of AGVs. According to the above literature [3-5], improving the utilization of AGVs is a feasible method to increase production, considering the urgency of time and manufacturing cost.

Improving the utilization of AGVs is equivalent to reducing waiting time, travelling time and blocking time to increase the working time of AGVs. Mohammad el at. [6] presented a modified ant colony algorithm to solve the conflict-free routing problem for AGVs. The goal was to improve the utilization of AGVs by reducing the blocking time. Ebben [7] studied four routing heuristics for battery management in an underground transportation system, which focused on the selection of the route to the battery station.

10.1016/j.procir.2019.03.155

According to the charging characteristics of valve-regulated lead-acid battery, the duration of each charging occurrence was reduced to get more working time.

These researches mentioned above provide good reference methods for improving the utilization of AGVs. However, most scholars only considered a single factor to improve the utilization of AGVs, and few researchers considered waiting time and travelling time together to increase the utilization of AGVs. Therefore, it is worth for further research. This paper attempts to fill the gap and propose two dual-stage charging strategies to improve the utilization of AGVs, thus achieving an increase in output.

The remaining parts are organized as follows. Section 2 illustrates the problem description and solution approaches. The case study and the results are presented in Section 3, while Section 4 is dedicated to the final discussion.

2. Problem description and solution approaches

2.1. Problem description

The considered manufacturing system is composed by S workstations. There are J AGVs use for products movement from one workstation to another. K ($K<J$) is the number of charging stations for AGVs automatic charging. The maximum daily production capacity of the manufacturing system is N. The manufacturing system needs to improve manufacturing capacity in a short term due to the rush orders occur or the product forecasts are underestimated. In order to solve this problem, it is planned to increase the maximum daily production capacity to M ($M>N$) by improving the utilization of AGVs without increasing the number of AGVs.

2.2. Solution approaches

In the proposed battery charging strategy, two stages including the selection of route and the selection of charging duration. The purpose of stage 1 is to reduce the routing time and travelling time of AGVs and two different routing heuristics are introduced to select the route of AGVs towards the charging station. In stage 2, the duration of charging time is selected to reduce the waiting time of the charging process. The details for the two stages are given in the following sections.

(1) Routing heuristics of AGVs for stage 1

Ebben [7] used some routing heuristics for the battery management of AGVs in an underground transportation system,which aimed at battery swapping rather than automatic charging. Since the layout of an underground transportation system is completely different from that of manufacturing system. The routing heuristics need to be modified appropriately. Two routing heuristics are described below in stage 1.

Selection of the nearest charging station (NCS): After the AGV had fininshed the job, the available charge is checked to determine if it is less than a pre-specified trigger level charge. When the AGV needs charging, it travels to the nearest charging station for automatic charging, which means the least travelling time. This is the most prevalent charging strategy used in the actual factory.

Selection of the minimum delay charging station (MDCS): When the amount of AGVs is greater than charging stations, it may happen that the AGVs are queued for charging at the charging stations. Such situation causes more waiting time to reduce the utilization of AGVs. The minimum delay charging station is selected in the this routing heuristics. The minimum delay charging station means that the AGV consumes the minimum amount of time considering waiting time in line and travelling time.

(2) Selecting charging duration for stage 2

The lithium-ion batteries own faster charging time and lighter weight than the valve-regulated lead acid batteries [8].The charging process of the lithium-ion battery can be divided into two stages, namely constant current charging and constant current voltage charging. As illustrated in Fig.1, charge time and battery capacity present a nonlinear relationship, the charging efficiency is higher in the initial stage [9].

AGVs will obtain more time to productive by reducing the duration of each charging occurrence. Therefore, in the case of the same number of AGVs, the factory would have the possibility of yielding more output.

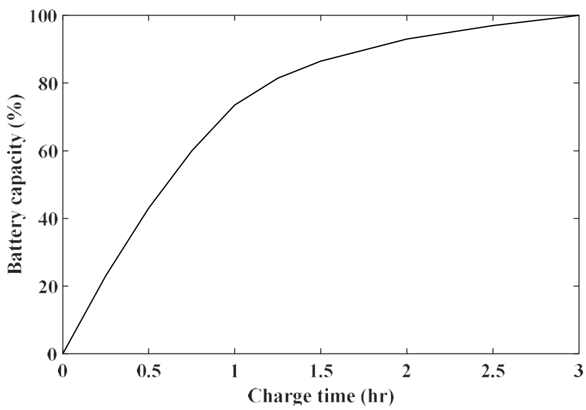

Fig. 1. Typical charging profile of a lithium-ion battery.

The time required for recharging depends on its depth of discharge (DoD), which refers to the percentage of battery discharge and its full capacity. Another important concept regarding the charging time for lithium-ion battery is the state of charge (SoC), which means the percentage of the remaining capacity of the battery and its full capacity. That is, DoD+SoC=100%. According to Mejdoubi el at. [10], the battery should not have more than 80% DoD.

In order to saving time, the charging duration was reduced to recharge the battery to a lower SoC level in the short term. According to Qazi el at. [11], 3 target SoCs were determined, namely 100% SoC, 95% SoC and 90% SoC.

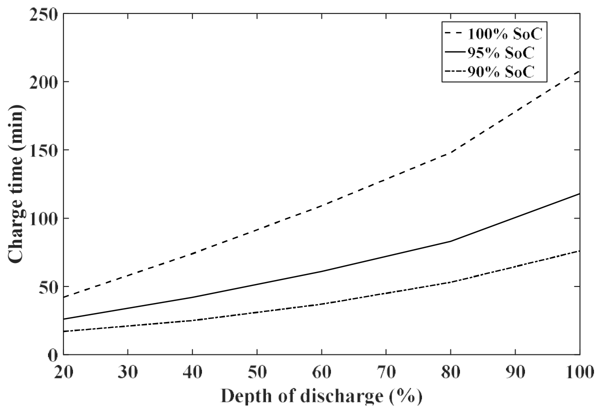

Fig. 2. Charge time and depth of discharge for a lithium-ion battery.

As shown in Fig. 2, relevant data are obtained from Hu el at. [12]. To further obtains the relationship between DoD and charge time, the regression technique is used to fit the data. According to the result, three fitted equations are got where x indicates DoD and y means charging time. Eqs. (1)-(3) represent the regression formulas for 100% SoC, 95% SoC, and 90% SoC respectively. Meanwhile, it can be found that the value of coefficient of determination (R^2) for all three equations is greater than 0.98, which indicates the precision of the results.

$$y = 31.365e^{0.3746x} \tag{1}$$

$$y = 19.055e^{0.3706x} \tag{2}$$

$$y = 11.809e^{0.3746x} \tag{3}$$

(3) The two dual-stage charing strategies

The two dual-stage charging strategies used in this study are described below.

Step 1: If the remaining charge of the empty AGV is less than the pre-specified trigger level charge, go to step 2. Otherwise, go to step 8.

Step 2: Select the battery charging strategy of NCS, go to step 3. Select the strategy of MDCS, go to step 4.

Step 3: The AGV moves to the nearest charging station from its current (outlet or loading point). If select 100% SoC, go to step 5, if select 95% SoC, go to step 5; otherwise, go to step 6.

Step 4: The AGV moves to the charging station with minimum time for waiting in line and travelling from its current location. If select 100% SoC, go to step 5, if select 95% SoC, go to step 5; otherwise, go to step 6.

Step 5: Set the charging duration to ensure the state of charge is 100% at the end of charging, then go to step 8.

Step 6: Set the charging duration to ensure the state of charge is 95% at the end of charging, then go to step 8.

Step 7: Set the charging duration to ensure the state of charge is 90% at the end of charging, then go to step 8.

Step 8: The AGV returns to the parking area and receive a new job.

The pre-specified trigger level charge in this study is 23% to ensure the residual capacity of the depleted battery goes not less than 20% after completing a new job. As shown in Fig. 3, the flow chart shows the two dual-stage charging strategies.

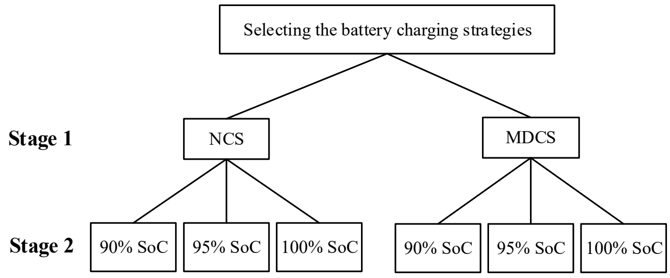

Fig. 3. The two dual-stage battery charging strategies.

3. Case study and results analysis

3.1. Layout of the manufacturing cell

As given in Fig.4, an actual manufacturing cell is studied in this research. Heat treatment of gearbox gears is implemented on this manufacturing cell. The direction of the path in the facility is unidirectional, which means that all AGVs can move in only one direction. There are 14 stations, which include 1 inlet, 1 outlet, 6 workstations, 5 charging stations and 1 parking area.

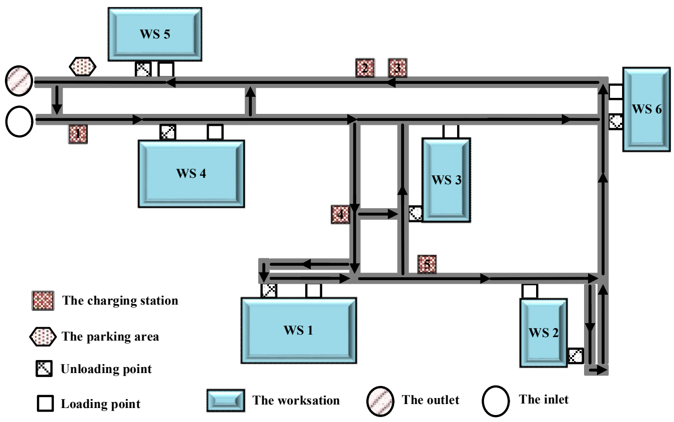

Fig. 4. Layout of a manufacturing cell.

The inlet used to produce the parts and then get released to different workstations; the parts are processed in the workstations; the outlet stores the finished parts after getting processed. At the battery stations, AGVs are capable of automatic charging; AGVs park and wait for missions in the parking area. Each workstation has a loading point and an unloading point, and their positions are different. Besides, only one AGV can be charged for each battery station at any time.

3.2. Processing time and routing time

In this study, 2 different types of parts names part A and part B are produced. Part A is generated every 90 seconds and part B is generated every 3 hours. At the time zero, part A is generated at the inlet. Workstation 1~5 are in a parallel relationship, and the part A only needs to be processed through any one the workstation 1~5. Part B can only be processed on the workstation 6. For example, part A is first produced at the inlet. After getting processed at the workstation 3, and finally send to the outlet for storage.

Table 1 shows the process times for two parts produced at the manufacturing facility.

Table1. Processing time in minutes with 6 workstations.

Part types	Part A	Part B
Workstation 1	20	-
Workstation 2	18	-
Workstation 3	17	-
Workstation 4	21	-
Workstation 5	22	-
Workstation 6	-	15

Table 2 shows the routing times that the loading point of workstations to charging stations.

Table 2. Routing times of the loading point to charging stations.

No.	Charging station				
	1	2	3	4	5
WS1	170	111	108	235	20
WS2	129	70	67	194	214
WS3	132	73	70	53	24
WS4	35	116	113	197	217
WS5	19	160	157	84	104
WS6	100	41	38	165	185

3.3. Assumptions and ampere consumption for different activities

Following assumptions are given for establishing the simulation model.

- The AGVs are identical;
- The breakdown of AGVs will not cause time loss;
- The driving speed of each AGV is 60 meters per minute;
- The time for each unloading task and loading task is 45 seconds;
- The charge losses due to deceleration and acceleration of AGVs are ignored;
- Each AGV is a unit load AGV.

Moreover, the factory operates 24 h per day to maximize the production of the system. All AGVs are working continuously.

The lithium-ion battery capacity for an AGV considered is 200ampere-hours. According to actual test, the amperes consumed by three different AGV activities are obtained. The blocking of AGVs consumes 5 amperes per hour; the travelling empty of AGVs consumes 10 amperes per hour and the travelling loaded of AGVs consumes 65 amperes per hour. The threshold level of battery represents the level of charge where the AGV is sent for automatic charging. According to Kaiser [13], the threshold is 80% DoD.

3.4. Number of AGVs

Using the two dual-stage charging strategies, variation in performance of the system in terms of total output is observed as the number of AGV is gradually increased. It is found that the number of AGV increased to 15, the output of the system decreased sharply. Hence, the upper limit for the number of AGV fix at 15 and the lower limit of the AGV stay at 4. Following Rossetti el at. [14] and Banks el at. [15], the period for each simulation experiment is determined to be 120 h and the number of replications for each simulation experiment is setted to be 50.

3.5. Results and analysis

The average utilization of AGVs with different number of AGVs using the two dual-stage charging strategies are shown in Fig.5 and Fig.6.

Table 3 shows the standard deviation and significance evaluation of the average utilization, which indicates the precision of the average utilization.

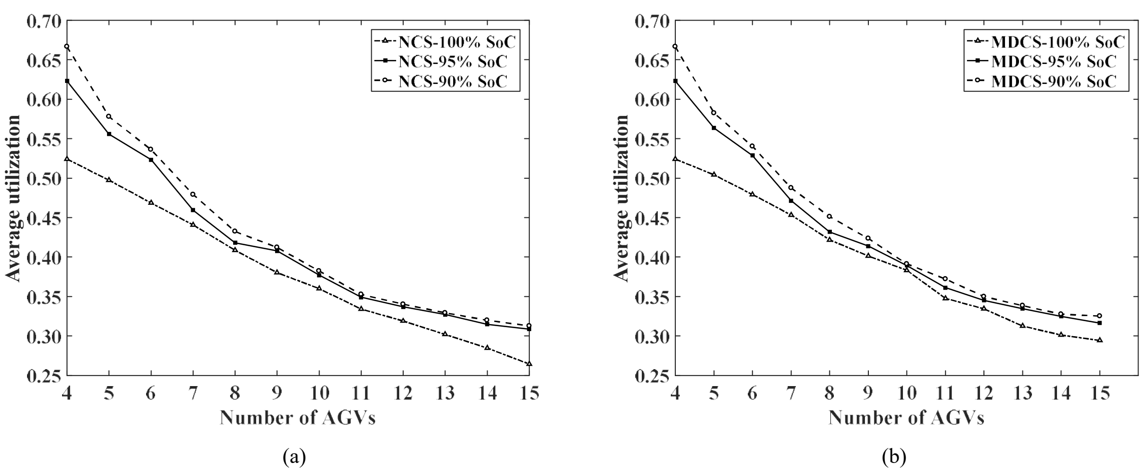

Fig. 5. Average utilization with numbers of AGVs and with different SoCs.

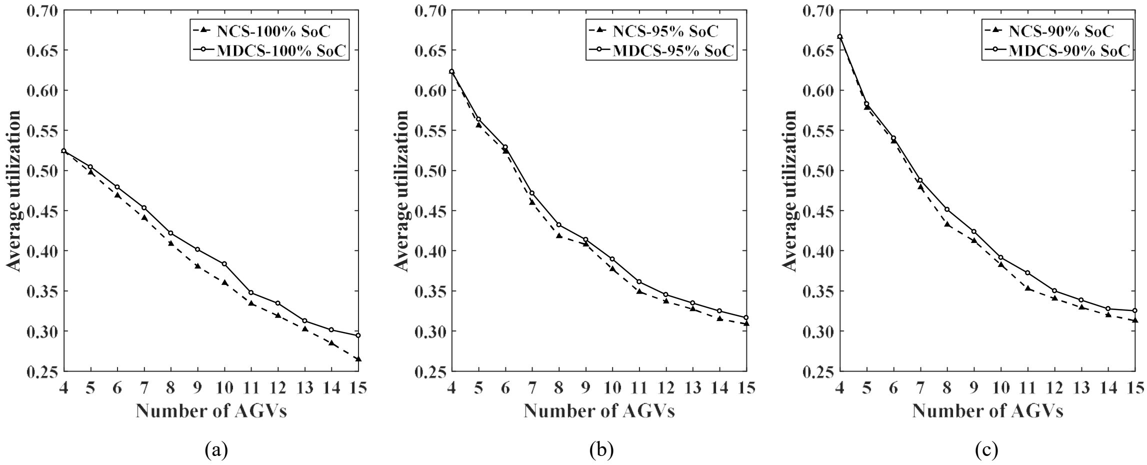

Fig. 6. Average utilization with number of AGVs and with different routings.

Table 3. The standard deviation and significance evaluation of the average utilization.

No	Routing heuristics	Mean value (%)			Standard deviation (%)			P-value		
		100%	95%	90%	100%	95%	90%	100%	95%	90%
4	NCS	52.41	62.31	66.66	0.28	0.32	0.27		P<0.01	
	MDCS	52.41	62.31	66.66	0.31	0.36	0.29			
5	NCS	49.73	55.59	57.78	0.35	0.41	0.34		P<0.01	
	MDCS	50.42	56.35	58.26	0.40	0.39	0.42			
6	NCS	46.86	52.33	53.61	0.40	0.32	0.47		P<0.01	
	MDCS	47.92	52.87	54.02	0.37	0.43	0.49			
7	NCS	44.07	45.94	47.92	0.40	0.32	0.43		P<0.01	
	MDCS	45.33	47.14	48.76	0.34	0.42	0.38			
8	NCS	40.86	41.81	43.23	0.35	0.34	0.29		P<0.01	
	MDCS	42.17	43.19	45.11	0.31	0.35	0.38			
9	NCS	38.02	40.77	41.21	0.40	0.32	0.40		P<0.01	
	MDCS	40.13	41.38	42.35	0.33	0.36	0.33			
10	NCS	35.99	37.69	38.24	0.39	0.36	0.38		P<0.01	
	MDCS	38.32	38.93	39.13	0.42	0.32	0.32			
11	NCS	33.41	34.89	35.26	0.28	0.27	0.33		P<0.01	
	MDCS	34.76	36.11	37.21	0.24	0.35	0.41			
12	NCS	31.89	33.68	34.02	0.29	0.29	0.33		P<0.01	
	MDCS	33.43	34.51	34.99	0.31	0.34	0.32			
13	NCS	30.19	32.72	32.93	0.37	0.25	0.27		P<0.01	
	MDCS	31.25	33.46	33.83	0.30	0.32	0.27			
14	NCS	28.47	31.48	31.98	0.25	0.30	0.27		P<0.01	
	MDCS	30.11	32.48	32.76	0.31	0.32	0.27			
15	NCS	26.44	30.87	31.27	0.29	0.25	0.26		P<0.01	
	MDCS	29.42	31.64	32.51	0.33	0.29	0.29			

As illustrated in Fig.5 (a) and (b), regarding to the same routing heuristics, decreasing the level of SoC helps improving the average utilization of AGVs.

According to Fig.6 (a) to (c), when the SoC-Level is the same, the dual-stage charging strategy of MDCS performance is better than the dual-stage charging strategy of NCS in terms of the average utilization of AGVs.

Once the number of AGVs is less than the number of charging stations, changing the routing heuristics cannot improve the utilization of AGVs.

More AGVs means less average utilization of AGVs. When the number of AGVs is less than the number of charging stations, the charging needs of the AGVs can be met. As the number of AGVs continues to increase, the system facing more and more congestion in addition to charging stations. A significant amount of blocked time occurs because of the congestion.

Table 4. Total output with the two dual-stage battery charging strategies.

No.	NCS			MDCS		
	100%	95%	90%	100%	95%	90%
4	148	208	228*	148	208	228*
5	211	242	261	214	245	265*
6	262	279	305	266	285	310*
7	275	292	315	285	299	322*
8	303	309	321	310	317	329*
9	323	325	330	328	331	335*
10	329	338	342	335	345	349*
11	340	347	350	345	352	355*
12	345	355	356	350	358	360*
13	348	352	358	352	355	361*
14	299	332	361	320	342	364*
15	247	314	335	299	330	345*

Note: *Largest among all the total output found from the two dual-stage battery charging strategies.

Table 4 shows the total output of the system with the two dual-stage battery charging strategies. It can be concluded that improving the average utilization of AGVs can increase the output of the system. It is also shown that the dual-stage charging strategy performs better than the single charging strategy, whicn only considers the routing heuristics.

According to the test results, the utilization of AGVs and the output of the system can be increased slightly by using the dual-stage charging strategy. This method can deal with small batches of rush orders. At present, the methods, which to increase the output of manufacturing system with AGVs, are difficult to complete in a short time and need a large investment. The method proposed in this paper can increase the production of the system in a short time without adding additional costs.

4. Conclusion

This paper proposes two dual-stage battery charging strategies for AGVs to improve the utilization of AGVs. These methods contribute to improving manufacturing capacity in a short term to meet the market demand. Results of this paper show that the charging strategy of MDCS performs better than the charging strategy of NCS in terms of the utilization of AGVs and the total output, which means that improving the utilization of AGVs helps to increasing the production of the system. And it should be noticed that even the increase in output is not very large, the factory should take such methods because changing the battery charging strategy does not require any additional investment.

The purpose of this study is to illustrate the dual-stage battery charging strategy effectively improve the utilization rather than maximized the utilization. In the future, optimization techniques can be used to solve this issue. Besides, the dual-stage battery charging strategy can be applied to other types of systems(e.g. container terminals, warehouse).

Acknowledgements

This research is supported by National Science and Technology Major Project of the Ministry of Science and Technology of China (2018ZX09201010-005) and The Intelligent Manufacturing project of the Ministry of Industry and Information Technology.

References

[1] Schmidt J, Claas M B, Matthias E, Lutz M K, Hans J A. Using battery-electric AGVs in container terminals — Assessing the potential and optimizing the economic viability. Research in Transportation Business & Management, 2015; 17: 99-111.

[2] Oliveira M, Galdames J P M, Vivaldini K T, Magalhães D V, Becker M. Battery state estimation for applications in intelligent warehouses. 2011; IEEE: Shanghai. 5511-5516.

[3] Zheng Y, Shigeru F. Simulation of multi-load AGV system based on JIT production environment. 2010; IEEE: Chengdu. 255-259.

[4] Kumar R, Prakash S S. A similarity score-based two-phase heuristic approach to solve the dynamic cellular facility layout for manufacturing systems. Engineering Optimization, 2017; 49(11): 1848-1867.

[5] Yang Z, Li C, Zhao Q. Dynamic time estimation based AGV dispatching algorithm in automated container terminal. 2018; IEEE: Wuhan. 7868-7873.

[6] Mohammad S M, Saeed D A, Farshid E, Vahid M. An ant colony algorithm (ACA) for solving the new integrated model of job shop scheduling and conflictfree routing of AGVs. Computers & Industrial Engineering, 2015; 86: 2-13.

[7] Ebben M J R . Logistic control in automated transportation networks. Production Engineering, 2001; 5(4): 373-382.

[8] Huang X, Zhu X, Xu K, Wu J. Research on power consumption of lithium ion battery protection circuit. Advanced Materials Research, 2014; 1049(4): 582-585.

[9] Xu J , Thomas H, Francis R, Wang J, Liang B. A review of processes and technologies for the recycling of lithium-ion secondary batteries. Journal of Power Sources, 2008; 177(2): 512-527.

[10] Mejdoubi A, Oukaour A, Chaoui H. State-of-charge and state-of-health lithium-ion batteries' diagnosis according to surface temperature variation. IEEE Transactions on Industrial Electronics, 2016; 63(4): 2391-2402.

[11] Qazi S K, Yoshinori S.Increasing manufacturing flexibility through battery management of automated guided vehicles. Computers & Industrial Engineering, 2018; 117: 225-236.

[12] Hu Y, Liu H, Liu P, Zhang Y. An analysis on the charging characteristics of lithium iron phosphate batteries for electric vehicles. Automotive Engineering, 2013; 35(4): 293-297.

[13] Kaiser R . Optimized battery-management system to improve storage lifetime in renewable energy systems. Journal of Power Sources, 2007; 168(1): 58-65.

[14] Banks J, Carson J, Nelson B, Nicol D. Discrete-event system simulation (4th ed.). 2005; Upper Saddle River, NJ: Pearson Education Inc.

[15] Rossetti M. Simulation modeling and arena. 2010; Hoboken, NJ: John Wiley and Sons Inc. 245–250.